Fowler (2024) on the Japanese commuting zones

2010 against 2020, at a cutoff of 0.977

August 26, 2026

What this deck reports

- Every table and figure of Fowler (2024) that the Japanese data supports, in the order that paper presents them.
- Delineations for 1980 to 2020; this deck shows the 2010 against 2020 comparison, the same decade spacing as the source paper.
- The wage-correlation block, that paper's Figure 2 and the Connection rows of its Table 2, waits on the Basic Survey on Wage Structure microdata.
- Full tables for all nine census years and both cutoffs are in the appendix.

Source paper: `https://doi.org/10.1038/s41597-024-03829-5`

The delineation

Methods section of Fowler (2024)

For municipalities i and j with commuting flows f_{ij} and resident labour forces W_i, the **proportional flow** is

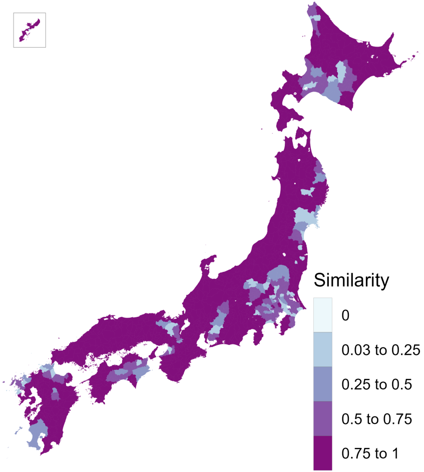
$$p_{ij} = \frac{f_{ij} + f_{ji}}{\min(W_i, W_j)}, \qquad p_{ij} \leq 0.999,$$

and the dissimilarity is $1 - p_{ij}$ with a zero diagonal.

- Average-linkage hierarchical clustering, cut at a fixed tree height of 0.977.
- Commuting matrix: residents mainly working, ordinary households, the only definition available for 2020.
- 1,678 municipalities: the four main islands and the Okinawa main island, plus every municipality joined to one by a permanent road link.
- Cores and urban areas for the fit statistics: the Urban Employment Area central cities and their areas.

Similarity between the 2010 and the 2020 delineations

Figure 1 of Fowler (2024)



Jaccard similarity: the overlap of a municipality's two commuting zones over their union. Mean 0.83.

Diagnostic statistics

Table 1 of Fowler (2024)

	2010	2020
Municipalities in the delineation	1,678	1,677
Commuting zones	235	227
Single-municipality zones	9	10
Municipalities in the largest zone	42	36
Non-contiguous zones	0	1
Smallest zone, resident labour force	530	475
Average zone, resident labour force	199,244	203,875
Largest zone, resident labour force	6,624,903	6,511,972
Smallest zone area (sq. km)	39	76
Average zone area (sq. km)	1,547	1,601
Largest zone area (sq. km)	5,744	5,268
Compactness of zones	0.30	0.30

Compactness is the Polsby–Popper ratio, $4\pi A/P^2$ of the dissolved zone, averaged across zones.

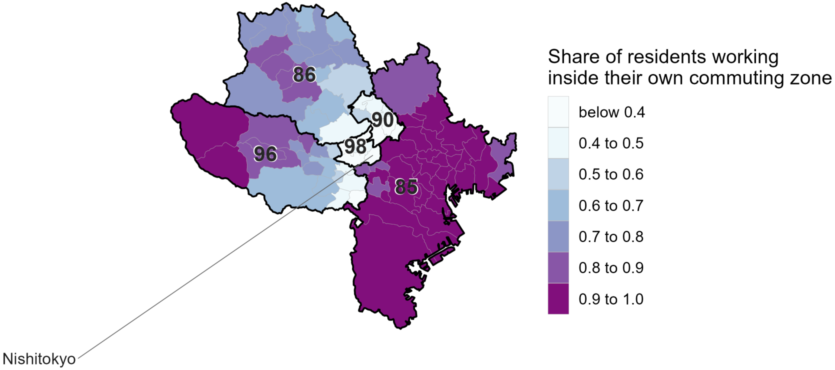
Fit statistics

Table 2 of Fowler (2024)

	2010	2020
Core		
Urban areas split across zones	31	33
Zones containing a core	71.5%	66.1%
Residents working in a core of their zone	44.9%	43.4%
Workforce living in a core of their zone	43.1%	42.2%
Connection		
Minimum wage correlation	pending	
Mean wage correlation	pending	
Mean wage correlation, multi-municipality zones	pending	
Containment		
Minimum contained	30.5%	35.8%
Mean contained	89.0%	89.5%
Share of the labour force contained	86.5%	86.2%

The commuting zone with the lowest mean containment

Figure 3 of Fowler (2024)



Five western Tokyo suburbs at 0.36, wedged between the Tokyo zone and the zones west of it.

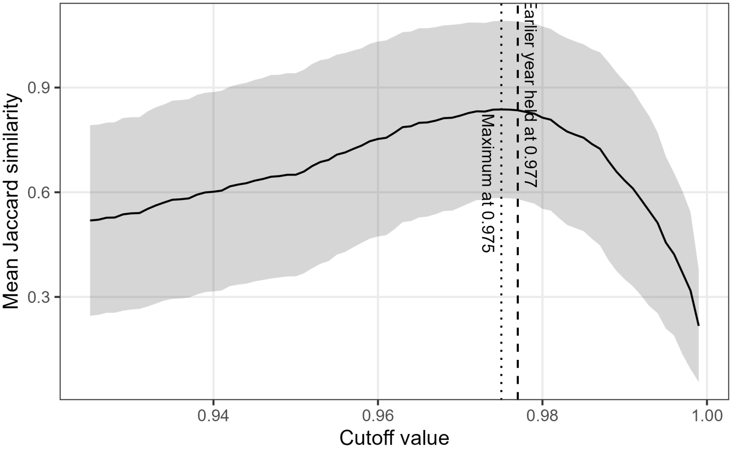
Technical validation

Technical Validation section of Fowler (2024)

- **The cutoff.** It is inherited rather than derived. Does the similarity between delineations pick out a better value?
- **Non-contiguity.** Zones that hold a municipality sharing its zone with none of its neighbours.
- **Split urban areas.** Externally defined urban areas cut across more than one commuting zone.

Similarity as a function of the cutoff

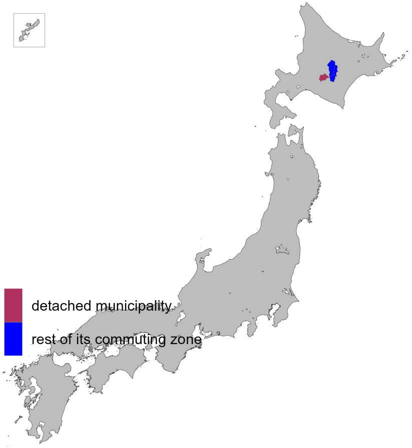
Figure 4 of Fowler (2024)



Holding 2010 at 0.977, the peak is 0.84 at 0.975, and the curve stays within 0.005 of it from 0.974 to 0.977, spanning 240 down to 227 zones.

Non-contiguous municipalities

Figure 5 of Fowler (2024)



One municipality in 2020, Shimukappu in Hokkaido, 991 metres from the nearest part of its own zone.

Reassignments that would remove non-contiguity

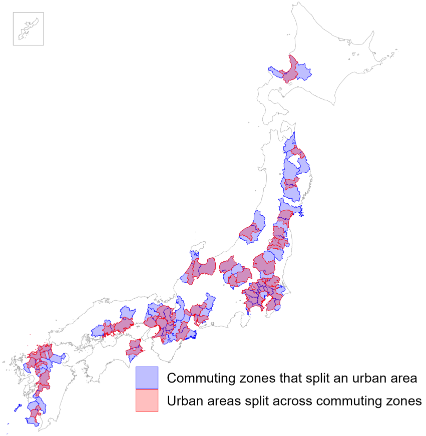
Table 3 of Fowler (2024)

Municipality	Zone	To own zone	Nearest zone	To that zone	Suggestion
Shimukappu	26	95.9%	18	2.9%	Retain

The assigned zone takes 96 percent of its commuters against 3 percent for the adjacent zone, so reassignment would fit the flows worse.

Urban areas split across commuting zones

Figure 6 of Fowler (2024)



33 of the 208 Urban Employment Areas in 2020, concentrated along the Pacific corridor.

Appendix: the full tables

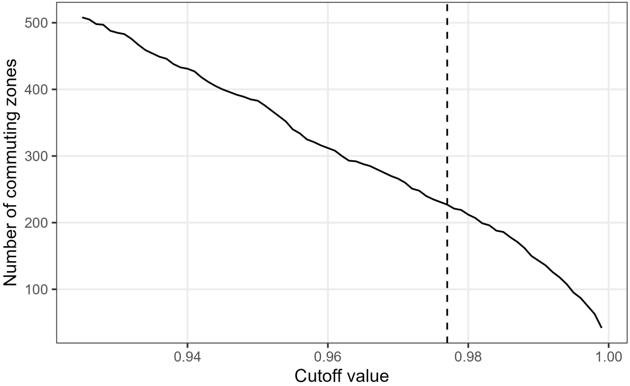
No counterpart in Fowler (2024)

Every table below covers all nine census years at both cutoff anchors. Links point at the branch `11-fowler-replication`.

- `validation/output/diagnostics_table.csv` — the diagnostic statistics, with both non-contiguity tests.
- `validation/output/containment_table.csv` — containment, including the labour-force weighted mean.
- `validation/output/core_table.csv` — the Core measures on the Urban Employment Area and on two densely inhabited district thresholds.
- `validation/output/reassignment_table.csv` — all 35 non-contiguity cases across the delineations.
- `validation/output/similarity_table.csv` — every comparison pair, including the pre-pandemic controls.
- `validation/output/cutoff_sweep.csv` — the sweep behind the similarity curve, with the diagnostic profile.

Appendix: the cutoff traded against the diagnostics

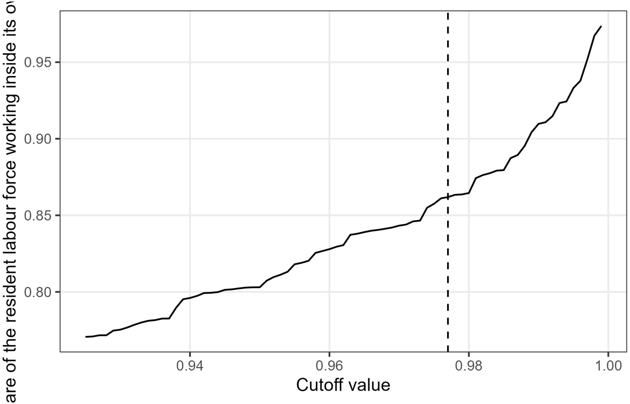
No counterpart in Fowler (2024)



`validation/output/figures/zone_count_vs_cutoff_2020.png`

Appendix: containment across the cutoff range

No counterpart in Fowler (2024)



`validation/output/figures/containment_vs_cutoff_2020.png`

Appendix: the other comparison pairs

No counterpart in Fowler (2024)

- The headline 2010 to 2020 comparison gives a mean similarity of 0.83, below both pre-pandemic controls.

- 2010 against 2015 gives 0.87 and 2005 against 2015 gives 0.85, at the same cutoff.

- Decade churn was larger before: 1980 against 1990 gives 0.76 and 1990 against 2000 gives 0.83.

- Curve for 2010 against 2015: `similarity_vs_cutoff_2010_2015_anchor0.977.png`
- Curve for 2005 against 2015: `similarity_vs_cutoff_2005_2015_anchor0.977.png`

Appendix: one departure from the existing pipeline

No counterpart in Fowler (2024)

The existing Japanese pipeline joins the flow file to its own transpose in a way that keeps a pair only when the flow from i to j is recorded, so a pair connected only in the reverse direction enters as unconnected.

- The matrix it clusters is asymmetric; this replication uses the symmetric numerator the source method specifies.
- Between 1.2 and 4.4 percent of pairs are affected in each year, moving up to four zones and up to 108 municipalities.
- Measured year by year in `validation/notes/method_crosscheck.md`.

Appendix: where everything lives

No counterpart in Fowler (2024)

- `validation/notes/replication_results.md` — the written results, with every number in this deck in context.
- `validation/notes/method_crosscheck.md` — the delineation formula across four implementations.
- `validation/notes/core_definition.md` — why the Urban Employment Area supplies the cores.
- `validation/code` — the nine scripts, run in the order they are numbered in the results note.